

WASP-2b

R-1723

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_130725 · created 2026-08-02T16:48:03+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456498.8494 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.047
Transit depth (Rp/Rs)^2	2.42 +/- 1.45 [%]
Orbital Inclination (inc)	85.05 +/- 2.87
Airmass coefficient 1 (a1)	0.354 +/- 0.09
Airmass coefficient 2 (a2)	0.9 +/- 0.95
Scatter in the residuals of the lightcurve fit is	26.7 %
Best Comparison Star	#3 - [261, 87]
Optimal Aperture	1.52
Optimal Annulus	13.17
Transit Duration (day)	0.065 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.047 vs 0.1326 (deviation 0.48σ)
Duration fit vs published	0.065 d vs 0.0737 d (deviation 0.27σ)
Mid-time O–C	-6.6 min
Depth SNR	3.3
Residual scatter	26.7 %

Light curve

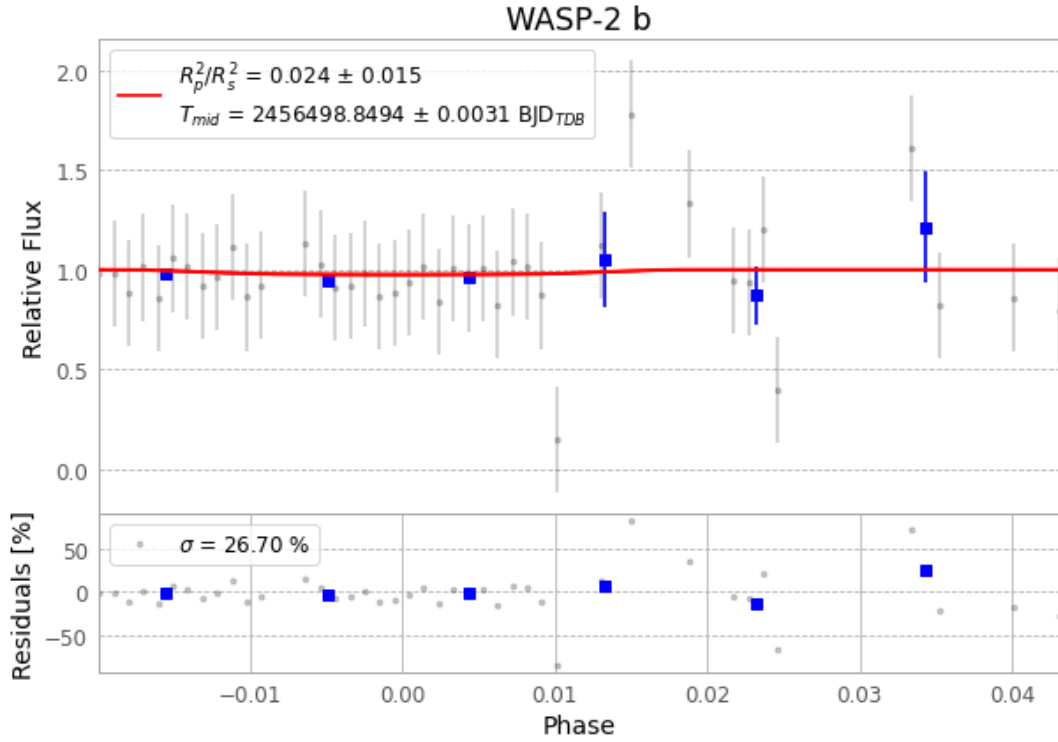


Figure 1 — FinalLightCurve_WASP-2 b_2013-07-25.png (SHA-256 45ea4651d4db10b8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [467, 294], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8c7168372c58c04e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-2130725071212.FITS → WASP-2130725103012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-130725.log (becc8d9881b8...), FinalLightCurve_WASP-2 b_2013-07-25.png (45ea4651d4db...), FinalLightCurve_WASP-2 b_2013-07-25.csv (f2a0d02961e2...)

Compute resources

Wall time 260.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 16:48Z.